

University of Debrecen, Faculty of Informatics

Logic in computer science



Zero-order logic (propositional logic)

sequent calculus

October 20, 2022

Gerhard Gentzen, 1934



Sequent

Let $A_1, A_2, \dots, A_n, B_1, B_2, \dots, B_m$ ($n, m \geq 0$) formulas in a zero-order logic language. The formula in the form

$$\top \wedge A_1 \wedge A_2 \wedge \dots \wedge A_n \supset B_1 \vee B_2 \vee \dots \vee B_m \vee \perp$$

is called a *sequent*, and it is shortly written as

$$A_1, A_2, \dots, A_n \rightarrow B_1, B_2, \dots, B_m \quad \text{or} \quad \Gamma \rightarrow \Delta,$$

where $\Gamma \Leftrightarrow \{A_1, A_2, \dots, A_n\}$ and $\Delta \Leftrightarrow \{B_1, B_2, \dots, B_m\}$.

Examples

$$\rightarrow \quad |\rightarrow| = \perp$$

$$\rightarrow B \quad |\rightarrow B| = |B|$$

$$A \rightarrow \quad |A \rightarrow| = |\neg A|$$

$$A \rightarrow B \quad |A \rightarrow B| = |A \supset B|$$

$$A_1, A_2 \rightarrow B_1, B_2 \quad |A_1, A_2 \rightarrow B_1, B_2| = |A_1 \wedge A_2 \supset B_1 \vee B_2|$$

Axiom schemes of Gentzen calculus

$$A, \Gamma \rightarrow \Delta, A$$

$$\Gamma \rightarrow \Delta, \top$$

$$\perp, \Gamma \rightarrow \Delta$$

Axiom schemes are tautologies.

Deduction rules of Gentzen calculus

$$(\rightarrow \equiv) \quad \frac{A, \Gamma \rightarrow \Delta, B; \quad B, \Gamma \rightarrow \Delta, A}{\Gamma \rightarrow \Delta, (A \equiv B)}$$

$$(\equiv \rightarrow) \quad \frac{A, B, \Gamma \rightarrow \Delta; \quad \Gamma \rightarrow \Delta, A, B}{(A \equiv B), \Gamma \rightarrow \Delta}$$

$$(\rightarrow \supset) \quad \frac{A, \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \supset B)}$$

$$(\supset \rightarrow) \quad \frac{\Gamma \rightarrow \Delta, A; \quad B, \Gamma \rightarrow \Delta}{(A \supset B), \Gamma \rightarrow \Delta}$$

$$(\rightarrow \wedge) \quad \frac{\Gamma \rightarrow \Delta, A; \quad \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \wedge B)}$$

$$(\wedge \rightarrow) \quad \frac{A, B, \Gamma \rightarrow \Delta}{(A \wedge B), \Gamma \rightarrow \Delta}$$

$$(\rightarrow \vee) \quad \frac{\Gamma \rightarrow \Delta, A, B}{\Gamma \rightarrow \Delta, (A \vee B)}$$

$$(\vee \rightarrow) \quad \frac{A, \Gamma \rightarrow \Delta; \quad B, \Gamma \rightarrow \Delta}{(A \vee B), \Gamma \rightarrow \Delta}$$

$$(\rightarrow \neg) \quad \frac{A, \Gamma \rightarrow \Delta}{\Gamma \rightarrow \Delta, \neg A}$$

$$(\neg \rightarrow) \quad \frac{\Gamma \rightarrow \Delta, A}{\neg A, \Gamma \rightarrow \Delta}$$

Deduction in sequent calculus

A sequent is provable in the Gentzen calculus, if

- the sequent is an axiom,
- or exists a deduction rule, where the sequent in question is written under the line, and the sequents on the top of the line are provable.

In other words, if we want to prove a sequent in the Gentzen calculus which is not an axiom, it is enough to find a deduction rule, where the sequent in question is written below the line, and prove the sequents above the line.

(If everything above the line is true, so is everything under the line.)

In the predicate logic, each sequent above the line of a deduction rule have lower degree, than the sequent under the line.

Soundness and completeness

- (a) The Gentzen sequent calculus is soundness, because when the sequent

$$A_1, A_2, \dots, A_n \rightarrow B_1, B_2, \dots, B_m$$

is provable in the Gentzen calculus, then the formula

$$\top \wedge A_1 \wedge A_2 \wedge \dots \wedge A_n \supset B_1 \vee B_2 \vee \dots \vee B_m \vee \perp$$

is valid.

- (b) The Gentzen sequent calculus is completeness, because if the formula A is a valid, then the sequent $\rightarrow A$ is provable in the Gentzen calculus.

University of Debrecen, Faculty of Informatics

Logic in computer science



Zero-order logic

(propositional logic)

deductions in sequent calculus

October 20, 2022

Semantic properties and valid sequents

- Γ is unsatisfiable if and only if $\Gamma \rightarrow$ is valid.
- $\Gamma \models A$ if and only if $\Gamma \rightarrow A$ is valid.
- $\models A$ (A is valid) if and only if $\rightarrow A$ is valid.
- $A \Leftrightarrow B$ if and only if $\rightarrow A \equiv B$ is valid.

Examples for deduction

$$A \equiv B \Leftrightarrow (A \supset B) \wedge (B \supset A)$$

$$A \supset B \Leftrightarrow \neg A \vee B$$

$$\neg\neg A \Leftrightarrow A$$

$$\neg(A \wedge B) \Leftrightarrow \neg A \vee \neg B$$

$$\neg(A \vee B) \Leftrightarrow \neg A \wedge \neg B$$

$$(A \wedge B) \vee C \Leftrightarrow (A \vee C) \wedge (B \vee C)$$

$$(A \vee B) \wedge C \Leftrightarrow (A \wedge C) \vee (B \wedge C)$$